WIGNER PHASE-SPACE REPRESENTATION AND COMPLEX SADDLE-POINT COALESCENCE IN THE TRANSVERSE WRONSKIAN OF RIEMANN'S Ξ-FUNCTION

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Abstract. We prove that the inner correlation integral $I(y,t)$ governing the transverse Wronskian $W(d,t) = \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2$ of Riemann's completed Ξ-function is proportional to the quantum-mechanical Wigner phase-space distribution $\mathcal{W}_\Phi(y,2t)$ of the Jacobi–Riemann kernel $\Phi(u)$, and that its 99.77% real-axis Fourier cancellation at $t=\gamma_1$ is eliminated by shifting the integration contour into the holomorphic strip $|\operatorname{Im}(x)| < \pi/4$. Our proof combines the holomorphic Jacobi differential identity for $\Phi(-u)=\Phi(u)$, double-exponential decay estimates on the vertical Cauchy walls $\xi = \pm L$, Hermitian reflection symmetry $G_y(-\xi+\mathrm{i}\eta) = \overline{G_y(\xi+\mathrm{i}\eta)}$ along horizontal contours, and a pitchfork saddle-point coalescence on the imaginary axis $\xi_* = 0$ for all $|y| \geq y_c(t)$. We further establish that Hudson's theorem and the zero-marginal identity $\int_{\mathbb{R}} I(y,\gamma_k)\,\mathrm{d}y = 2|\Xi(\gamma_k)|^2 = 0$ force $I(y,\gamma_k)$ to change sign in y, while the monotone hyperbolic moment weight $y\sinh(dy)$ amplifies the positive coalesced tail.

1. Introduction

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ denote Riemann's completed zeta function, and set $s = \frac{1}{2} + d + \mathrm{i}t$ with transverse offset $d = \sigma - \frac{1}{2} \in \mathbb{R}$ and spectral height $t \in \mathbb{R}$. Writing $z = -\mathrm{i}(s-\frac{1}{2}) = t - \mathrm{i}d$, Riemann's entire function $\Xi(z) := \xi(\frac{1}{2}+\mathrm{i}z) = R(d,t) + \mathrm{i}I_{\mathrm{im}}(d,t)$ admits the Fourier–Laplace integral representation [11, 10, 13]

$$\Xi(t-\mathrm{i}d) = 2\int_0^\infty \Phi(u)\cosh(du)\cos(tu)\,\mathrm{d}u = \int_{-\infty}^{\infty} \Phi(u)\,\mathrm{e}^{du-\mathrm{i}tu}\,\mathrm{d}u, \tag{1.1}$$

where the rapidly decaying kernel $\Phi(u)$ is generated by the classic Jacobi theta series $\theta(x) = \sum_{n=-\infty}^{\infty}\mathrm{e}^{-\pi n^2 x}$ through

$$\Phi(u) := \sum_{n=1}^{\infty}\phi_n(u), \qquad \phi_n(u) := \Big(2\pi^2 n^4\mathrm{e}^{9u} - 3\pi n^2\mathrm{e}^{5u}\Big)\exp\Big(-\pi n^2\mathrm{e}^{4u}\Big). \tag{1.2}$$

By the positivity criteria of Speiser [12], Hinkkanen [5], and Lagarias [8], the absence of off-critical zeros of $\Xi(t-\mathrm{i}d)$ in $d>0$ is intimately linked to the sign of the real part of the logarithmic derivative $\operatorname{Re}(\xi'(s)/\xi(s)) = W(d,t)/|\Xi(t-\mathrm{i}d)|^2$, where the *transverse Wronskian* is defined by

$$W(d,t) := \frac{1}{2}\frac{\partial}{\partial d}\big|\Xi(t-\mathrm{i}d)\big|^2 = R\,\partial_d R + I_{\mathrm{im}}\,\partial_d I_{\mathrm{im}} = R\,\partial_t I_{\mathrm{im}} - I_{\mathrm{im}}\,\partial_t R. \tag{1.3}$$

When (1.3) is expanded directly as a double integral over $(u,v)\in\mathbb{R}^2$ along the real axis, numerical and analytical evaluation is obstructed by severe oscillatory cancellation: at the first non-trivial zero $t=\gamma_1 \approx 14.13472514$, positive and negative cosine lobes cancel to 99.7697%. The purpose of this article is to resolve this cancellation problem analytically by establishing the exact phase-space and complex-contour structure of $W(d,t)$.

Theorem 1.1 (Wigner Phase-Space Identity and Pitchfork Saddle Coalescence)**.** *Let $d>0$ and $t>0$, and define the mirror coordinates $x=u+v$ and $y=u-v$ with mirror convolution kernel $G_y(x) := \Phi\Big(\frac{x+y}{2}\Big)\,\Phi\Big(\frac{x-y}{2}\Big)$. Then the following properties hold:*

Date: October 2026.

2020 *Mathematics Subject Classification.* Primary 11M06, 30E15; Secondary 81S30, 42B10, 65D30.

Key words and phrases. Riemann Xi-function, Jacobi theta kernel, transverse Wronskian, Wigner phase-space distribution, Hudson theorem, complex saddle-point coalescence, steepest descent.

(i) ***Wigner–Hudson phase-space representation and zero-marginal law:*** *The transverse Wronskian and its parity-reduced counterpart* $\widetilde{W}(d,t) := W(d,t)/d$ *admit the exact representations*

$$W(d,t) = \frac{1}{2}\int_0^\infty y\sinh(dy)\, I(y,t)\,\mathrm{d}y, \qquad \widetilde{W}(d,t) = \frac{1}{2}\int_0^\infty y^2\,\frac{\sinh(dy)}{dy}\, I(y,t)\,\mathrm{d}y, \tag{1.4}$$

where the inner correlation integral $I(y,t) := \int_{-\infty}^{\infty} G_y(x)\cos(tx)\,\mathrm{d}x$ *is proportional to the quantum-mechanical Wigner distribution of* Φ*:*

$$I(y,t) = 2\pi\,\mathcal{W}_\Phi(y,2t), \qquad \mathcal{W}_\Phi(y,\omega) := \frac{1}{2\pi}\int_{-\infty}^{\infty}\Phi\!\left(\frac{y+x}{2}\right)\Phi\!\left(\frac{y-x}{2}\right)\mathrm{e}^{\mathrm{i}(\omega/2)x}\,\mathrm{d}x. \tag{1.5}$$

At every real zero γ_k *of* Ξ*, the marginal integral vanishes identically,* $\int_{-\infty}^{\infty} I(y,\gamma_k)\,\mathrm{d}y = 2|\Xi(\gamma_k)|^2 = 0$*, and since* Φ *is non-Gaussian, Hudson's theorem guarantees that* $I(y,t)$ *changes sign in the* (y,t)*-plane.*

(ii) ***Holomorphic strip, vertical wall decay, and Hermitian contour symmetry:*** *For each fixed* $y \in \mathbb{R}$*, the function* $z \mapsto G_y(z)$ *extends holomorphically (without non-holomorphic absolute values) to the open strip* $\mathcal{S}_{\pi/4} := \{z = \xi + \mathrm{i}\eta \in \mathbb{C} : |\eta| < \pi/4\}$ *and satisfies the even parity* $G_y(-z) = G_y(z)$*. On any vertical segment* $\xi = \pm L$ *with* $|\eta| < \pi/4$*,* $G_y(\pm L + \mathrm{i}\eta)$ *decays double-exponentially as* $\mathcal{O}(\mathrm{e}^{9L}\exp(-2\pi\cosh(2y)\mathrm{e}^{2L}\cos(2\eta)))$*, whereas* $|\eta| = \pi/4$ *forms a natural boundary where* $\cos(2\eta) = 0$*. Consequently, for every* $\eta \in (0,\pi/4)$ *and* $\omega > 0$*,*

$$I(y,\omega) = \mathrm{e}^{-\omega\eta}\int_{-\infty}^{\infty}\mathrm{Re}\Big[G_y(\xi+\mathrm{i}\eta)\,\mathrm{e}^{\mathrm{i}\omega\xi}\Big]\mathrm{d}\xi, \tag{1.6}$$

and G_y *obeys the exact Hermitian symmetry* $G_y(-\xi+\mathrm{i}\eta) = \overline{G_y(\xi+\mathrm{i}\eta)}$*, taking real non-negative values on the imaginary axis:* $G_y(\mathrm{i}\eta) = |\Phi((y+\mathrm{i}\eta)/2)|^2 > 0$ *whenever* $\Phi((y+\mathrm{i}\eta)/2) \neq 0$*.*

(iii) ***Pitchfork saddle-point coalescence:*** *For each fixed frequency* $\omega > 0$*, the complex saddle points* $x_* \in \mathcal{S}_{\pi/4}$ *solving* $\partial_x \ln G_y(x) + \mathrm{i}\omega = 0$ *undergo a pitchfork bifurcation at a critical mirror shift* $y_c(\omega) > 0$ *determined by the simultaneous system*

$$\mathrm{Im}\big[\partial_x \ln G_y(\mathrm{i}\eta_c)\big] + \omega = 0, \qquad \partial_\xi^2 \ln G_y(\mathrm{i}\eta_c) = 0.$$

For $0 \le |y| < y_c(\omega)$*, the dominant saddle splits into a conjugate-reflected off-axis pair* $x_*^{\pm}(y,\omega) = \pm\xi_*(y,\omega) + \mathrm{i}\eta_*(y,\omega)$ *with* $\xi_* > 0$ *and* $0 < \eta_* < \pi/4$*. For all* $|y| \ge y_c(\omega)$*, the pair coalesces onto the imaginary axis at* $x_*(y,\omega) = \mathrm{i}\eta_*(y,\omega)$ *(*$\xi_* \equiv 0$*) with strictly negative horizontal Hessian* $\partial_\xi^2 \ln G_y(\mathrm{i}\eta_*) < 0$*, eliminating oscillatory cancellation along the horizontal steepest-descent contour.*

To fix the quantitative scale of Theorem 1.1, we collect the canonical threshold constants and first-zero benchmark values (verified to 50–60 decimal places in Section 5) in a single display:

$$\begin{aligned}
\eta_{\max} &= \frac{\pi}{4} \approx 0.7853981634, & \gamma_1 &\approx 14.1347251417,\ \ \omega_1 = 2\gamma_1 \approx 28.2694502835,\\
y_c(\omega_1) &\approx 0.4320205303, & \eta_*(y_c,\omega_1) &\approx 0.6027150239,\\
y_c^{\mathrm{inner}}(\omega_1) &= \frac{1}{2}\ln\!\left(\frac{\omega_1}{4\pi}\right) \approx 0.40537874, & \rho_{\mathrm{real}}(\omega_1) &= \frac{I_+}{|I_-|}\Big|_{\eta=0} \approx 1.002308,\\
\rho_{\mathrm{lift}}(\omega_1) &= \frac{I_+}{|I_-|}\Big|_{\eta=0.50} \approx 2547.9547, & \rho_{\mathrm{coal}}(\gamma_1) &= \frac{I_+}{|I_-|}\Big|_{\eta=0.18} \approx 1.547\times 10^{11}.
\end{aligned} \tag{1.7}$$

Table 1 summarizes the distinction between the classical foundations in analytic number theory and quantum harmonic analysis and the specific structural results proven in the present work.

1.1. **Proof overview.** The proof of Theorem 1.1 is organized into three quantitative and structural ingredients:

TABLE 1. Separation between classical foundations in the literature and the novel analytical results established in this work.

Classical Author(s)	**Established Classical Result**	**Novel Contribution in This Work**
Riemann (1859) [11], Pólya (1926) [10]	Representation of $\Xi(z)$ via the Jacobi-theta kernel $\Phi(u)$ and Fourier convolution integrals.	Transformation of $\partial_d\|\Xi\|^2$ into (x,y) mirror coordinates and identification of $I(y,t)=2\pi\mathcal{W}_\Phi(y,2t)$.
Speiser (1934) [12], Hinkkanen (1997) [5], Lagarias (1999) [8]	Equivalence between zero-free regions in $d>0$ and positivity of $\mathrm{Re}(\xi'(s)/\xi(s))=W(d,t)/\|\Xi\|^2$.	Parity-reduced hyperbolic-sinc representation $\widetilde{W}(d,t)=W(d,t)/d$ and removal of the $d\to 0$ degeneracy.
Hudson (1974) [6], Wigner (1932) [14]	A Wigner distribution $\mathcal{W}_\psi(q,p)$ on $\mathbb{R}^2$ is everywhere non-negative if and only if ψ is Gaussian.	Proof that non-Gaussianity of $\Phi(u)$ and $\int_{\mathbb{R}} I(y,\gamma_k)\,\mathrm{d}y = 0$ force sign changes of $I(y,t)$ in the (y,t)-plane.
Karlin (1968) [7], Csordas–Norfolk–Varga (1986) [2]	Total positivity of kernels and Turán inequalities for Jensen polynomials of Ξ.	Monotone hyperbolic moment lever $y\sinh(dy)-y_0\sinh(dy_0)$ matching the single sign-change point $y_0(\gamma_k)$ of $I(y,\gamma_k)$.

(1) *First structural ingredient (Mirror factorization and Wigner zero-marginal identity, Section 2):* Using the holomorphic evenness $\Phi(-v)=\Phi(v)$ and the linear change of variables $(x=u+v, y=u-v)$ with Jacobian $\frac{1}{2}$, we factor the double integral for $|\Xi(t-\mathrm{i}d)|^2$ into the longitudinal Fourier transform $I(y,t)=2\pi\mathcal{W}_\Phi(y,2t)$ and the transverse hyperbolic weight $\cosh(dy)$:

$$\big|\Xi(t-\mathrm{i}d)\big|^2=\frac{1}{2}\int_{-\infty}^{\infty}\cosh(dy)\,I(y,t)\,\mathrm{d}y \implies \int_{-\infty}^{\infty} I(y,\gamma_k)\,\mathrm{d}y = 2\big|\Xi(\gamma_k)\big|^2=0. \tag{1.8}$$

(2) *Second quantitative ingredient (Cauchy contour box and Hermitian reflection in $|\eta|<\pi/4$, Section 3):* By replacing $\cos(\omega x)$ with $\mathrm{Re}(\mathrm{e}^{\mathrm{i}\omega x})$ on the real axis prior to contour deformation and bounding the vertical segments $\xi=\pm L$ for $0\le\tau\le\eta<\pi/4$, we prove the uniform side-wall decay estimate

$$\big|G_y(\pm L+\mathrm{i}\tau)\,\mathrm{e}^{\mathrm{i}\omega(\pm L+\mathrm{i}\tau)}\big|\le C_0\,\mathrm{e}^{9L-\omega\tau}\exp\Big(-2\pi\cosh(2y)\,\mathrm{e}^{2L}\cos(2\eta)\Big)\xrightarrow{L\to\infty}0, \tag{1.9}$$

together with the exact Hermitian symmetry $G_y(-\xi+\mathrm{i}\eta)=\overline{G_y(\xi+\mathrm{i}\eta)}$.

(3) *Third quantitative ingredient (Pitchfork bifurcation and Hessian negativity on the imaginary axis, Section 4):* Restricting $\ln G_y(\xi+\mathrm{i}\eta)$ to the imaginary axis $\xi=0$ where $G_y(\mathrm{i}\eta)=|\Phi(\frac{y+\mathrm{i}\eta}{2})|^2>0$, the Cauchy–Riemann equations yield

$$\partial_\xi \ln G_y(\mathrm{i}\eta)=-\mathrm{i}\,\partial_\eta \ln G_y(\mathrm{i}\eta),\qquad \partial_\xi^2 \ln G_y(\mathrm{i}\eta)=-\partial_\eta^2 \ln G_y(\mathrm{i}\eta)=\frac{1}{2}\,\mathrm{Re}\left[\frac{\mathrm{d}^2}{\mathrm{d}u^2}\ln\Phi\left(\frac{y+\mathrm{i}\eta}{2}\right)\right]. \tag{1.10}$$

When $|y|\ge y_c(\omega)$, the stationary-phase equation $\partial_\eta \ln G_y(\mathrm{i}\eta_*)=\omega$ admits a unique steepest-descent solution on the lower branch $\eta_*(y,\omega)\in(0,\eta_c]$ where $\partial_\xi^2 \ln G_y(\mathrm{i}\eta_*)<0$.

Combining ingredients (1), (2), and (3) yields Theorem 1.1 directly: ingredient (1) establishes the Wigner representation (1.4)–(1.5) and the zero-marginal constraint (1.8); ingredient (2) justifies the contour shift (1.6) across the open strip $|\eta|<\pi/4$ and proves the natural boundary at $|\eta|=\pi/4$; and ingredient (3) establishes the pitchfork coalescence of $x_*^{\pm}(y,\omega)$ at $y_c(\omega)$ and the elimination of real-axis oscillatory cancellation.

1.2. **Notation and conventions.** Throughout this paper, $s=\sigma+\mathrm{i}t=\frac{1}{2}+d+\mathrm{i}t$ denotes a complex variable with $d=\sigma-\frac{1}{2}\in\mathbb{R}$ and $t\in\mathbb{R}$, and $z=t-\mathrm{i}d$. The completed Riemann function $\Xi(z)$ is normalized by (1.1) so that $\Xi(0)=\xi(1/2)\approx 0.4971207782$. Note that if one works in the rescaled variable $u_{\mathrm{scaled}}=2u$ (as in Section 5 when comparing $\omega=2t$ against $\omega=t$), replacing $\cos(tx)$ by $\cos(2tx)$ simply rescales the frequency parameter from $\omega=t$ to $\omega=2t$; we state all saddle-point and contour formulas in terms of a general positive frequency parameter $\omega>0$ so that both conventions are covered simultaneously without ambiguity. For complex

$w \in \mathbb{C}$, $\overline{w}$ denotes complex conjugation, and $|w| = (w\overline{w})^{1/2}$ is reserved strictly for moduli; we never substitute $|u|$ into the argument of $\Phi(u)$. Vinogradov's notation $A \ll B$ and Landau's $A = \mathcal{O}(B)$ mean that $|A| \le CB$ for an absolute constant $C > 0$ independent of L, ξ, y, d, and t, unless a subscript (such as $\ll_\eta$) explicitly indicates dependence on a parameter.

2. Mirror coordinates and the Wigner phase-space identity

We begin by deriving the exact mirror-coordinate factorization of $|\Xi(t - \mathrm{i}d)|^2$ and $W(d,t)$, justifying all integral interchanges via Fubini's theorem, and connecting the inner kernel to quantum phase-space analysis.

Proposition 2.1 (Fubini justification and mirror factorization)**.** *For all $(d,t) \in \mathbb{R}^2$, the double integral representation of $|\Xi(t - \mathrm{i}d)|^2$ converges absolutely and satisfies*

$$\big|\Xi(t - \mathrm{i}d)\big|^2 = \frac{1}{2}\int_{-\infty}^{\infty} \cosh(dy)\, I(y,t)\,\mathrm{d}y = \int_0^{\infty} \cosh(dy)\, I(y,t)\,\mathrm{d}y, \tag{2.1}$$

where the inner correlation integral $I(y,t)$ is an even, smooth function of both y and t given by

$$\boxed{I(y,t) = \int_{-\infty}^{\infty} \Phi\Big(\frac{x+y}{2}\Big)\,\Phi\Big(\frac{x-y}{2}\Big)\cos(tx)\,\mathrm{d}x = 2\pi\,\mathcal{W}_\Phi(y,2t).} \tag{2.2}$$

Proof. For real $u \in \mathbb{R}$, the Jacobi-theta kernel $\Phi(u)$ is smooth, strictly positive, even ($\Phi(-u) = \Phi(u)$, proven holomorphically in Lemma 3.1 below), and satisfies the super-exponential decay bound $0 < \Phi(u) \le C_1 \mathrm{e}^{9|u|}\exp(-\pi \mathrm{e}^{4|u|})$. Consequently, for any fixed $d \in \mathbb{R}$ and $k \ge 0$, the function $u \mapsto |u|^k \Phi(u)\mathrm{e}^{|d||u|}$ belongs to $L^1(\mathbb{R})$. Writing $\Xi(t - \mathrm{i}d) = \int_{\mathbb{R}} \Phi(u)\mathrm{e}^{du - \mathrm{i}tu}\,\mathrm{d}u$ and $\overline{\Xi(t - \mathrm{i}d)} = \int_{\mathbb{R}} \Phi(v)\mathrm{e}^{dv + \mathrm{i}tv}\,\mathrm{d}v$, Tonelli's theorem ensures that the product $\Phi(u)\Phi(v)\mathrm{e}^{d(u+v)}$ is integrable over $\mathbb{R}^2$. By Fubini's theorem,

$$\big|\Xi(t - \mathrm{i}d)\big|^2 = \iint_{\mathbb{R}^2} \Phi(u)\Phi(v)\,\mathrm{e}^{d(u+v)}\cos\big(t(u-v)\big)\,\mathrm{d}u\,\mathrm{d}v.$$

Performing the reflection $v \mapsto -v$ and invoking the even parity $\Phi(-v) = \Phi(v)$ yields

$$\begin{aligned}\big|\Xi(t - \mathrm{i}d)\big|^2 &= \iint_{\mathbb{R}^2} \Phi(u)\Phi(v)\,\mathrm{e}^{d(u-v)}\cos\big(t(u+v)\big)\,\mathrm{d}u\,\mathrm{d}v \\ &= \iint_{\mathbb{R}^2} \Phi(u)\Phi(v)\cosh\big(d(u-v)\big)\cos\big(t(u+v)\big)\,\mathrm{d}u\,\mathrm{d}v,\end{aligned}$$

where the second equality follows by averaging the integrand with its image under $(u,v) \mapsto (v,u)$. Now introduce the linear mirror transformation $x = u + v$, $y = u - v$, with inverse $u = \frac{x+y}{2}$, $v = \frac{x-y}{2}$ and Jacobian determinant $|\frac{\partial(u,v)}{\partial(x,y)}| = \frac{1}{2}$. Since $\Phi(\frac{x+y}{2})\Phi(\frac{x-y}{2})\cosh(dy)$ is strictly positive and integrable on $\mathbb{R}^2$, Fubini's theorem permits separating the x- and y-integrals to obtain (2.1) and the first equality in (2.2). Finally, since Φ is even and real on $\mathbb{R}$, $G_y(-x) = G_y(x)$ implies $\int_{\mathbb{R}} G_y(x)\sin(tx)\,\mathrm{d}x = 0$, so replacing $\cos(tx)$ by $\mathrm{e}^{\mathrm{i}tx}$ and comparing with the definition of $\mathcal{W}_\Phi(y,2t)$ in (1.5) completes the proof. □

Corollary 2.2 (Differentiation under the integral sign and parity reduction)**.** *For all $(d,t) \in \mathbb{R}^2$, the transverse Wronskian $W(d,t) = \frac{1}{2}\partial_d|\Xi(t - \mathrm{i}d)|^2$ is an odd analytic function of d, i.e., $W(-d,t) = -W(d,t)$, so all of its even transverse derivatives vanish identically on the critical line: $\partial_d^{2m} W(0,t) \equiv 0$ for all $m \ge 0$. Dividing by d yields the smooth, even parity-reduced Wronskian*

$$\widetilde{W}(d,t) := \frac{W(d,t)}{d} = \frac{1}{2}\int_0^{\infty} y^2\,\frac{\sinh(dy)}{dy}\,I(y,t)\,\mathrm{d}y, \qquad \widetilde{W}(0,t) = \big(\Xi'(t)\big)^2 - \Xi(t)\Xi''(t). \tag{2.3}$$

Proof. For any compact interval $|d| \le D$, the differentiated integrand $|y\sinh(dy)I(y,t)| \le |y|\sinh(D|y|)I(y,0)$ is dominated by an $L^1(\mathbb{R})$ function because $I(y,0) \ll \mathrm{e}^{9|y|}\exp(-2\pi\mathrm{e}^{2|y|})$. Lebesgue's dominated convergence theorem therefore justifies differentiating (2.1) under the integral sign with respect to d arbitrarily many times, giving (1.4) and (2.3). Evaluating at $d = 0$ via $\lim_{d\to 0}\frac{\sinh(dy)}{dy} = 1$ and using $\partial_d^2|\Xi(t - \mathrm{i}d)|^2\big|_{d=0} = 2[(\Xi'(t))^2 - \Xi(t)\Xi''(t)]$ yields the boundary identity in (2.3). □

Proposition 2.3 (Hudson zero-marginal theorem and Karlin hyperbolic moment lever)**.** *Let $\gamma_k \in \mathbb{R}$ be any real zero of $\Xi(\gamma_k) = 0$. Then:*

(a) *The marginal integral of $I(y, \gamma_k)$ vanishes identically:*

$$\int_{-\infty}^{\infty} I(y, \gamma_k)\, \mathrm{d}y = 2 \int_0^{\infty} I(y, \gamma_k)\, \mathrm{d}y = 2\big|\Xi(\gamma_k)\big|^2 = 0. \tag{2.4}$$

Since $I(0, \gamma_k) = \int_{\mathbb{R}} \Phi(x/2)^2 \cos(\gamma_k x)\, \mathrm{d}x \neq 0$ and $I(\cdot, \gamma_k)$ is continuous, $I(y, \gamma_k)$ must assume both negative and positive values on $y \in (0, \infty)$, in accordance with Hudson's theorem [6] for the non-Gaussian wavepacket Φ.

(b) *For any reference point $y_0 > 0$ and any $d > 0$, the transverse Wronskian satisfies the exact lever identity*

$$W(d, t) = \frac{1}{2} y_0 \sinh(d y_0) \big|\Xi(t)\big|^2 + \frac{1}{2} \int_0^{\infty} \Big(y \sinh(dy) - y_0 \sinh(d y_0) \Big) I(y, t)\, \mathrm{d}y. \tag{2.5}$$

In particular, whenever $I(y, t)$ has a single sign change on $(0, \infty)$ from $I(y, t) \leq 0$ on $(0, y_0)$ to $I(y, t) \geq 0$ on (y_0, ∞) (not identically zero), the strict monotonicity of $y \mapsto y \sinh(dy)$ on $(0, \infty)$ implies that $(y \sinh(dy) - y_0 \sinh(d y_0)) I(y, t) \geq 0$ for all $y \geq 0$, whence $W(d, t) > 0$ for all $d > 0$.

Proof. Setting $d = 0$ in (2.1) gives $\int_0^{\infty} I(y, t)\, \mathrm{d}y = |\Xi(t)|^2$, which vanishes at $t = \gamma_k$, proving (2.4). Multiplying $\int_0^{\infty} I(y, t)\, \mathrm{d}y = |\Xi(t)|^2$ by $\frac{1}{2} y_0 \sinh(d y_0)$ and subtracting from (1.4) produces (2.5). Since $w_d(y) := y \sinh(dy)$ has derivative $w_d'(y) = \sinh(dy) + dy \cosh(dy) > 0$ for all $d > 0$ and $y > 0$, $w_d(y) - w_d(y_0)$ is negative on $[0, y_0)$ and positive on (y_0, ∞). If $I(y, t)$ changes sign from negative on $(0, y_0)$ to positive on (y_0, ∞), both factors have the same sign everywhere on $[0, \infty)$, so the integrand in (2.5) is non-negative and strictly positive on an open set, yielding $W(d, t) > 0$. □

3. Holomorphic continuation and the Cauchy contour box in $|\operatorname{Im}(x)| < \pi/4$

To eliminate the oscillatory cancellation in $I(y, \omega) = \int_{\mathbb{R}} G_y(x) \cos(\omega x)\, \mathrm{d}x$, we deform the line of integration into the complex plane $x = \xi + \mathrm{i}\eta$. A rigorous complex-analytic treatment requires avoiding non-holomorphic operations such as $u \mapsto |u|$ and verifying absolute convergence and vertical wall bounds prior to shifting the contour.

Lemma 3.1 (Holomorphic continuation and even parity of $\Phi(u)$ in $|\operatorname{Im}(u)| < \pi/8$)**.** *The series (1.2) defines a holomorphic function $\Phi(u)$ on the open horizontal strip*

$$\mathcal{S}_{\pi/8}^{(u)} := \left\{ u = \sigma + \mathrm{i}\tau \in \mathbb{C} : |\tau| < \frac{\pi}{8} \right\}, \tag{3.1}$$

and satisfies the exact holomorphic evenness identity $\Phi(-u) = \Phi(u)$ for all $u \in \mathcal{S}_{\pi/8}^{(u)}$.

Proof. For $u = \sigma + \mathrm{i}\tau \in \mathcal{S}_{\pi/8}^{(u)}$, we have $|4\tau| < \pi/2$, so $\cos(4\tau) > 0$ and $\operatorname{Re}(\mathrm{e}^{4u}) = \mathrm{e}^{4\sigma} \cos(4\tau) > 0$. On any compact subset $K \subset \mathcal{S}_{\pi/8}^{(u)}$, there exist constants $\sigma_{\min} \leq \sigma \leq \sigma_{\max}$ and $\delta_K := \min_K \cos(4\tau) > 0$ such that

$$|\phi_n(u)| \leq \Big(2\pi^2 n^4 \mathrm{e}^{9\sigma_{\max}} + 3\pi n^2 \mathrm{e}^{5\sigma_{\max}} \Big) \exp\Big(-\pi n^2 \mathrm{e}^{4\sigma_{\min}} \delta_K \Big).$$

By the Weierstrass M-test, the series $\sum_{n=1}^{\infty} \phi_n(u)$ converges absolutely and uniformly on K, so $\Phi(u)$ is holomorphic on $\mathcal{S}_{\pi/8}^{(u)}$. Furthermore, for $\operatorname{Re}(\psi) > 0$, Jacobi's theta function $\theta(\psi) = 1 + 2\sum_{n=1}^{\infty} \mathrm{e}^{-\pi n^2 \psi}$ is holomorphic and obeys Poisson summation $\theta(\psi^{-1}) = \psi^{1/2} \theta(\psi)$. Setting $\psi = \mathrm{e}^{4u}$ with $|\operatorname{Im}(u)| < \pi/8$ ensures $\operatorname{Re}(\mathrm{e}^{\pm 4u}) > 0$, so $\theta(\mathrm{e}^{-4u}) = \mathrm{e}^{2u} \theta(\mathrm{e}^{4u})$, or equivalently $f(u) := \mathrm{e}^{u} \theta(\mathrm{e}^{4u}) = \mathrm{e}^{-u} \theta(\mathrm{e}^{-4u}) = f(-u)$. Applying the second-order differential operator $\frac{1}{16}(\partial_u^2 - 1)$ term-by-term to $f(u) = \mathrm{e}^{u} + 2\sum_{n=1}^{\infty} \mathrm{e}^{u} \mathrm{e}^{-\pi n^2 \mathrm{e}^{4u}}$ annihilates the $n = 0$ term $(\partial_u^2 - 1)\mathrm{e}^{u} = 0$ and produces

$$\Phi(u) = \frac{1}{16} \left(\frac{\mathrm{d}^2}{\mathrm{d}u^2} - 1 \right) \Big[\mathrm{e}^{u}\, \theta(\mathrm{e}^{4u}) \Big] = \frac{1}{16} \left(\frac{\mathrm{d}^2}{\mathrm{d}u^2} - 1 \right) f(u). \tag{3.2}$$

Since $f(-u) = f(u)$ and $\partial_u^2 - 1$ is invariant under $u \mapsto -u$, it follows identically that $\Phi(-u) = \Phi(u)$ for all $u \in \mathcal{S}_{\pi/8}^{(u)}$. □

3.1. **Pre-move justification and double-exponential vertical wall bounds.** Before shifting the contour of $I(y,\omega) = \int_{-\infty}^{\infty} G_y(x)\cos(\omega x)\,\mathrm{d}x$, we note that one cannot deform $\cos(\omega x) = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\omega x} + \mathrm{e}^{-\mathrm{i}\omega x})$ directly into the upper half-plane $\mathrm{Im}(x) = \eta > 0$, because the component $\mathrm{e}^{-\mathrm{i}\omega(\xi+\mathrm{i}\eta)} = \mathrm{e}^{+\omega\eta}\mathrm{e}^{-\mathrm{i}\omega\xi}$ grows exponentially with η while $G_y(\xi + \mathrm{i}\eta)$ remains bounded of order $\mathcal{O}(1)$ near $\xi = 0$. Instead, we first exploit the even symmetry $G_y(-x) = G_y(x)$ on the real axis $\eta = 0$: since $G_y(x)\sin(\omega x)$ is odd and belongs to $L^1(\mathbb{R})$, its integral over $\mathbb{R}$ vanishes identically, allowing the pre-move replacement

$$I(y,\omega) = \int_{-\infty}^{\infty} G_y(x)\cos(\omega x)\,\mathrm{d}x = \mathrm{Re}\left(\lim_{L\to\infty}\int_{-L}^{L} G_y(x)\,\mathrm{e}^{\mathrm{i}\omega x}\,\mathrm{d}x\right). \tag{3.3}$$

We now apply Cauchy's integral theorem to the holomorphic integrand $F_{y,\omega}(z) := G_y(z)\,\mathrm{e}^{\mathrm{i}\omega z}$ around the closed rectangular contour $\partial\mathcal{R}_{L,\eta}$ with vertices $-L, +L, +L + \mathrm{i}\eta, -L + \mathrm{i}\eta$, where $L > |y| + 1$ and $0 < \eta < \pi/4$. Since $z = \xi + \mathrm{i}\tau \in \mathcal{R}_{L,\eta}$ implies $|\,\mathrm{Im}(\frac{z\pm y}{2})| = \frac{\tau}{2} \le \frac{\eta}{2} < \frac{\pi}{8}$, Lemma 3.1 guarantees that $F_{y,\omega}(z)$ is holomorphic throughout an open neighborhood of $\mathcal{R}_{L,\eta}$, so $\oint_{\partial\mathcal{R}_{L,\eta}} F_{y,\omega}(z)\,\mathrm{d}z = 0$.

Proposition 3.2 (Vertical wall decay and natural boundary at $|\eta| = \pi/4$)**.** *Fix $y \in \mathbb{R}$, $\omega > 0$, and $0 < \eta < \pi/4$. For every $L \ge |y| + 1$ and $\tau \in [0,\eta]$, the convolution kernel on the vertical segments $z = \pm L + \mathrm{i}\tau$ obeys the uniform double-exponential bound*

$$|G_y(\pm L + \mathrm{i}\tau)| \le C_2\,\mathrm{e}^{9L}\exp\Big(-2\pi\cosh(2y)\,\mathrm{e}^{2L}\cos(2\eta)\Big), \tag{3.4}$$

where $C_2 = 16\pi^4$. Consequently, the vertical side-wall integrals $\mathcal{J}_\pm(L,\eta) := \int_0^\eta F_{y,\omega}(\pm L + \mathrm{i}\tau)\,\mathrm{i}\,\mathrm{d}\tau$ satisfy $|\mathcal{J}_\pm(L,\eta)| \to 0$ double-exponentially as $L \to \infty$. Conversely, on the boundary lines $\eta = \pm\pi/4$, $\cos(2\eta) = \cos(\pi/2) = 0$, so $|\exp(-\pi n^2\mathrm{e}^{2(\xi\pm y)}\mathrm{e}^{\pm\mathrm{i}\pi/2})| = 1$ for all $n \ge 1$ and the series (1.2) diverges everywhere on $\mathrm{Im}(z) = \pm\pi/4$.

Proof. By the holomorphic evenness $G_y(-z) = G_y(z)$, it suffices to bound $z = +L + \mathrm{i}\tau$ with $L \ge |y| + 1$ and $\tau \in [0,\eta]$. Here both arguments $u_\pm = \frac{L\pm y + \mathrm{i}\tau}{2}$ satisfy $\mathrm{Re}(u_\pm) = \frac{L\pm y}{2} \ge \frac{1}{2} > 0$ and $\cos(4\,\mathrm{Im}(u_\pm)) = \cos(2\tau) \ge \cos(2\eta) > 0$. Since $2\pi^2 n^4\mathrm{e}^{9\sigma} + 3\pi n^2\mathrm{e}^{5\sigma} \le 3\pi^2 n^4\mathrm{e}^{9\sigma}$ for $\sigma \ge 1/2$, we have

$$|\Phi(u_\pm)| \le 3\pi^2\mathrm{e}^{9(L\pm y)/2}\sum_{n=1}^{\infty} n^4\mathrm{e}^{-\pi n^2\mathrm{e}^{2(L\pm y)}\cos(2\eta)} \le 4\pi^2\mathrm{e}^{9(L\pm y)/2}\exp\Big(-\pi\mathrm{e}^{2(L\pm y)}\cos(2\eta)\Big),$$

where the tail $n \ge 2$ is bounded by Lemma A.1 in Appendix A. Multiplying the bounds for $\Phi(u_+)$ and $\Phi(u_-)$ and noting that $\mathrm{e}^{2(L+y)} + \mathrm{e}^{2(L-y)} = 2\cosh(2y)\mathrm{e}^{2L}$ yields (3.4). Integrating over $\tau \in [0,\eta]$ with $|\mathrm{e}^{\mathrm{i}\omega(\pm L+\mathrm{i}\tau)}| = \mathrm{e}^{-\omega\tau} \le 1$ gives $|\mathcal{J}_\pm(L,\eta)| \le \eta C_2\mathrm{e}^{9L}\exp(-2\pi\cosh(2y)\mathrm{e}^{2L}\cos(2\eta)) \to 0$ as $L \to \infty$. At $\eta = \pi/4$, $\cos(2\eta) = 0$, so the Gaussian factor $\exp(-\pi n^2\mathrm{e}^{2(\xi\pm y)}\cos(2\eta))$ has unit modulus and the terms of (1.2) grow as $n^4 \to \infty$. □

The bookkeeping of the four segments of the rectangular contour $\partial\mathcal{R}_{L,\eta}$ is summarized below:

Contour Segment	**Parameterization**	**Asymptotic Bound as $L \to \infty$**	**Limit**
Bottom Γ_{bot}	$z = \xi \in [-L, L]$	$\int_{-L}^{L} G_y(\xi)\mathrm{e}^{\mathrm{i}\omega\xi}\,\mathrm{d}\xi + \mathcal{O}(\mathrm{e}^{-2\pi\cosh(2y)\mathrm{e}^{2L}})$	$+I(y,\omega)$
Right wall Γ_{right}	$z = L + \mathrm{i}\tau$, $\tau \in [0,\eta]$	$\le \eta C_2\mathrm{e}^{9L}\exp(-2\pi\cosh(2y)\mathrm{e}^{2L}\cos(2\eta))$	0
Top wall Γ_{top}	$z = \xi + \mathrm{i}\eta$, $\xi \in [L, -L]$	$-\mathrm{e}^{-\omega\eta}\int_{-L}^{L} G_y(\xi + \mathrm{i}\eta)\mathrm{e}^{\mathrm{i}\omega\xi}\,\mathrm{d}\xi$	$-I(y,\omega)$
Left wall Γ_{left}	$z = -L + \mathrm{i}\tau$, $\tau \in [\eta, 0]$	$\le \eta C_2\mathrm{e}^{9L}\exp(-2\pi\cosh(2y)\mathrm{e}^{2L}\cos(2\eta))$	0

3.2. **Hermitian contour symmetry and imaginary-axis positivity.**

Proposition 3.3 (Hermitian contour symmetry)**.** *For every $y \in \mathbb{R}$ and every $\xi + \mathrm{i}\eta \in \mathcal{S}_{\pi/4}$, the convolution kernel G_y satisfies*

$$G_y(-\xi + \mathrm{i}\eta) = \overline{G_y(\xi + \mathrm{i}\eta)}, \qquad G_y(\mathrm{i}\eta) = \left|\Phi\left(\frac{y + \mathrm{i}\eta}{2}\right)\right|^2 \ge 0. \tag{3.5}$$

Consequently, $\operatorname{Re} G_y(\xi+\mathrm{i}\eta)$ *is an even function of* ξ, $\operatorname{Im} G_y(\xi+\mathrm{i}\eta)$ *is an odd function of* ξ, *and the shifted representation* (1.6) *reduces to the one-sided integral*

$$I(y,\omega) = 2\mathrm{e}^{-\omega\eta}\int_0^\infty \Big[\operatorname{Re} G_y(\xi+\mathrm{i}\eta)\cos(\omega\xi) - \operatorname{Im} G_y(\xi+\mathrm{i}\eta)\sin(\omega\xi)\Big]\mathrm{d}\xi. \tag{3.6}$$

Proof. Since $\Phi(u)$ is real-valued for real u, the Schwarz reflection principle gives $\Phi(\overline{u}) = \overline{\Phi(u)}$ for all $u \in \mathcal{S}^{(u)}_{\pi/8}$. Combined with Lemma 3.1 ($\Phi(-u) = \Phi(u)$), this implies $G_y(\overline{z}) = \overline{G_y(z)}$ and $G_y(-z) = G_y(z)$. Therefore, for $z = \xi + \mathrm{i}\eta \in \mathcal{S}_{\pi/4}$,

$$G_y(-\xi+\mathrm{i}\eta) = G_y\big(-(\xi-\mathrm{i}\eta)\big) = G_y(\xi-\mathrm{i}\eta) = G_y(\overline{z}) = \overline{G_y(\xi+\mathrm{i}\eta)}.$$

Evaluating at $\xi = 0$ and using $\Phi(\frac{-y+\mathrm{i}\eta}{2}) = \Phi(-\frac{y-\mathrm{i}\eta}{2}) = \Phi(\frac{y-\mathrm{i}\eta}{2}) = \overline{\Phi(\frac{y+\mathrm{i}\eta}{2})}$ yields $G_y(\mathrm{i}\eta) = |\Phi(\frac{y+\mathrm{i}\eta}{2})|^2 \geq 0$. Since $\operatorname{Re}[G_y(\xi+\mathrm{i}\eta)\mathrm{e}^{\mathrm{i}\omega\xi}] = \operatorname{Re} G_y(\xi+\mathrm{i}\eta)\cos(\omega\xi) - \operatorname{Im} G_y(\xi+\mathrm{i}\eta)\sin(\omega\xi)$ is the difference of two products of same-parity functions of ξ, it is an even function of ξ, which gives (3.6). □

4. Pitchfork saddle-point coalescence and elimination of real-axis cancellation

We now analyze the complex saddle points of the holomorphic phase function $\Psi_{y,\omega}(x) := \ln G_y(x) + \mathrm{i}\omega x$ inside the strip $\mathcal{S}_{\pi/4}$. Differentiating with respect to x yields the exact stationary-phase equation

$$\Psi'_{y,\omega}(x) = \frac{\partial}{\partial x}\ln G_y(x) + \mathrm{i}\omega = \frac{1}{2}\left[\frac{\Phi'}{\Phi}\left(\frac{x+y}{2}\right) + \frac{\Phi'}{\Phi}\left(\frac{x-y}{2}\right)\right] + \mathrm{i}\omega = 0. \tag{4.1}$$

4.1. **Asymptotic two-regime structure and pitchfork bifurcation.** To understand why the roots of (4.1) transition from an off-axis pair to a single imaginary-axis saddle as $|y|$ increases, we examine the two natural asymptotic regimes of $G_y(x) = \Phi(\frac{x+y}{2})\Phi(\frac{x-y}{2})$:

Proposition 4.1 (Outer two-saddle regime vs. inner coalesced regime)**.** *Let* $\omega > 0$ *and* $y \geq 0$.

(a) ***Outer regime*** *(*$\xi > y \geq 0$*):* *When* $\operatorname{Re}(x) = \xi > y$, *both arguments* $u_\pm = \frac{x\pm y}{2}$ *have positive real part* $\frac{\xi\pm y}{2} > 0$. *Retaining the leading* $n = 1$ *mode* $\phi_1(u_\pm) \approx 2\pi^2\mathrm{e}^{9u_\pm}\exp(-\pi\mathrm{e}^{4u_\pm})$ *gives* $G_y(x) \approx 4\pi^4\mathrm{e}^{9x}\mathrm{e}^{-2\pi\cosh(2y)\mathrm{e}^{2x}}$, *whose stationary-phase condition* $4\pi\cosh(2y)\mathrm{e}^{2x_*} = 9 + \mathrm{i}\omega$ *yields the conjugate-reflected off-axis pair* $x_*^\pm = \pm\xi_*^{\mathrm{out}}(y,\omega) + \mathrm{i}\eta_*^{\mathrm{out}}(\omega)$ *with*

$$\xi_*^{\mathrm{out}}(y,\omega) = \frac{1}{2}\ln\left(\frac{\sqrt{\omega^2+81}}{4\pi\cosh(2y)}\right), \qquad \eta_*^{\mathrm{out}}(\omega) = \frac{1}{2}\arctan\left(\frac{\omega}{9}\right) = \frac{\pi}{4} - \frac{1}{2}\arctan\left(\frac{9}{\omega}\right) < \frac{\pi}{4}. \tag{4.2}$$

(b) ***Inner reflected regime*** *(*$|\xi| < y$*):* *Near the imaginary axis (*$|\xi| < y$*), invoking the holomorphic parity* $\Phi(\frac{x-y}{2}) = \Phi(\frac{y-x}{2})$ *ensures that both arguments* $v_\pm = \frac{y\pm x}{2}$ *have strictly positive real part* $\frac{y\pm\xi}{2} > 0$. *The leading* $n = 1$ *approximation then takes the even-in-x form*

$$G_y(x) = \Phi\left(\frac{y+x}{2}\right)\Phi\left(\frac{y-x}{2}\right) \approx 4\pi^4\mathrm{e}^{9y}\exp\Big(-2\pi\mathrm{e}^{2y}\cosh(2x)\Big), \tag{4.3}$$

whose stationary-phase equation $-4\pi\mathrm{e}^{2y}\sinh(2x_*) + \mathrm{i}\omega = 0$, *i.e.,* $\sinh(2\xi_*)\cos(2\eta_*) + \mathrm{i}\cosh(2\xi_*)\sin(2\eta_*) = \mathrm{i}\frac{\omega}{4\pi\mathrm{e}^{2y}}$, *undergoes an exact pitchfork bifurcation at* $y_c^{\mathrm{inner}}(\omega) = \frac{1}{2}\ln(\frac{\omega}{4\pi})$: *for* $y \geq y_c^{\mathrm{inner}}(\omega)$, *the saddle lies purely on the imaginary axis at* $\xi_* = 0$ *with height*

$$\eta_*^{\mathrm{inner}}(y,\omega) = \frac{1}{2}\arcsin\left(\frac{\omega}{4\pi\mathrm{e}^{2y}}\right) \in \left(0, \frac{\pi}{4}\right]. \tag{4.4}$$

Proof. In part (a), differentiating $\ln(4\pi^4) + 9x - 2\pi\cosh(2y)\mathrm{e}^{2x} + \mathrm{i}\omega x$ with respect to x gives $9 - 4\pi\cosh(2y)\mathrm{e}^{2\xi_*+\mathrm{i}2\eta_*} + \mathrm{i}\omega = 0$. Equating real and imaginary parts yields $4\pi\cosh(2y)\mathrm{e}^{2\xi_*}\cos(2\eta_*) = 9$ and $4\pi\cosh(2y)\mathrm{e}^{2\xi_*}\sin(2\eta_*) = \omega$, from which (4.2) follows by division and taking moduli. By Proposition 3.3, $\overline{\Psi'_{y,\omega}(x)} = -\Psi'_{y,\omega}(-\overline{x})$, so x_*^+ is a root of (4.1) if and only if $x_*^- = -\overline{x_*^+} = -\xi_* + \mathrm{i}\eta_*$ is a root. In part (b), for $|\xi| < y$, replacing $\Phi(\frac{x-y}{2})$ by $\Phi(\frac{y-x}{2})$ via Lemma 3.1 gives

$\frac{9(y+x)}{2} + \frac{9(y-x)}{2} = 9y$ and $\mathrm{e}^{2(y+x)} + \mathrm{e}^{2(y-x)} = 2\mathrm{e}^{2y}\cosh(2x)$, establishing (4.3). For $y \geq \frac{1}{2}\ln(\frac{\omega}{4\pi})$, $\frac{\omega}{4\pi\mathrm{e}^{2y}} \leq 1$, so $\xi_* = 0$ and $\sin(2\eta_*) = \frac{\omega}{4\pi\mathrm{e}^{2y}}$ solve $\sinh(2x_*) = \mathrm{i}\frac{\omega}{4\pi\mathrm{e}^{2y}}$ uniquely in $\eta_* \in (0, \pi/4]$. □

We now prove the exact pitchfork bifurcation theorem for the full kernel $G_y(x)$ without any $n = 1$ truncation.

Theorem 4.2 (Exact pitchfork bifurcation on the imaginary axis)**.** *Along the imaginary axis $x = \mathrm{i}\eta$ with $\eta \in (0, \pi/4)$ and $y > 0$ outside the zeros of $\Phi(\frac{y+\mathrm{i}\eta}{2})$, the real log-amplitude $L_y(\eta) := \ln G_y(\mathrm{i}\eta) = 2\ln\left|\Phi\left(\frac{y+\mathrm{i}\eta}{2}\right)\right|$ satisfies the Cauchy–Riemann identities*

$$\frac{\partial}{\partial x}\ln G_y(\mathrm{i}\eta) = -\mathrm{i}L_y'(\eta) = \mathrm{i}\,\mathrm{Im}\left[\frac{\Phi'}{\Phi}\left(\frac{y+\mathrm{i}\eta}{2}\right)\right], \qquad \frac{\partial^2}{\partial\xi^2}\ln G_y(\xi+\mathrm{i}\eta)\Big|_{\xi=0} = -L_y''(\eta). \tag{4.5}$$

Consequently:

(i) *A point $x_* = \mathrm{i}\eta_*$ on the imaginary axis is a stationary point of $\Psi_{y,\omega}(x)$ if and only if $L_y'(\eta_*) = \omega$, and the horizontal contour $\xi \mapsto \xi + \mathrm{i}\eta_*$ is a local steepest-descent maximum of $\mathrm{Re}\,\Psi_{y,\omega}(\xi+\mathrm{i}\eta_*)$ at $\xi = 0$ if and only if $\partial_\xi^2 \ln G_y(\mathrm{i}\eta_*) = -L_y''(\eta_*) < 0$ (i.e., $L_y''(\eta_*) > 0$).*

(ii) *Near any bifurcation point (y_c, η_c) where $L_{y_c}'(\eta_c) = \omega$, $L_{y_c}''(\eta_c) = 0$, $\partial_y L_{y_c}'(\eta_c) > 0$, and $L_{y_c}'''(\eta_c) < 0$, the local holomorphic Taylor expansion of $\ln G_y(\mathrm{i}\eta_c + \Delta z)$ in $\Delta z = \xi + \mathrm{i}(\eta - \eta_c)$ around $\Delta z = 0$ takes the normal form*

$$\ln G_y(\mathrm{i}\eta_c + \Delta z) = L_y(\eta_c) - \mathrm{i}L_y'(\eta_c)\,\Delta z - \frac{1}{2}L_y''(\eta_c)\,(\Delta z)^2 + \frac{\mathrm{i}}{6}L_y'''(\eta_c)\,(\Delta z)^3 + \frac{a_4}{24}(\Delta z)^4 + \mathcal{O}((\Delta z)^5), \tag{4.6}$$

where $a_4 := \partial_z^4 \ln G_{y_c}(\mathrm{i}\eta_c) = L_{y_c}^{(4)}(\eta_c)$. Setting $\Psi_{y,\omega}'(\mathrm{i}\eta_c + \Delta z) = 0$ yields the quadratic coalescence law

$$(\Delta z)^2 = \big(\xi_* + \mathrm{i}(\eta_* - \eta_c)\big)^2 = K^2\,(y_c - y) + \mathcal{O}(|y - y_c|^{3/2}), \qquad K := \sqrt{\frac{2\,\partial_y L_{y_c}'(\eta_c)}{-L_{y_c}'''(\eta_c)}} > 0, \tag{4.7}$$

which splits into two conjugate-reflected off-axis saddles $\xi_*^{\pm}(y) = \pm K\sqrt{y_c - y} + \mathcal{O}(y_c - y)$ for $y < y_c$ and into the imaginary-axis steepest-descent branch $\xi_* \equiv 0$, $\eta_*(y) = \eta_c - K\sqrt{y - y_c} + \mathcal{O}(y - y_c)$ with $\partial_\xi^2 \ln G_y(\mathrm{i}\eta_*) = -L_y''(\eta_*) < 0$ for $y > y_c$.

Proof. By Proposition 3.3, $\ln G_y(-\xi + \mathrm{i}\eta) = \overline{\ln G_y(\xi + \mathrm{i}\eta)}$. Expanding the holomorphic function $z \mapsto \ln G_y(z)$ in powers of $\Delta z = z - \mathrm{i}\eta_c = \xi + \mathrm{i}(\eta - \eta_c)$ around $z = \mathrm{i}\eta_c$ and noting that $\partial_z = \partial_\xi = -\mathrm{i}\partial_\eta$ gives $\partial_z^k \ln G_y(\mathrm{i}\eta_c) = (-\mathrm{i})^k L_y^{(k)}(\eta_c)$, which yields both identities in (4.5) and the expansion (4.6). Differentiating (4.6) with respect to Δz at (y, η_c) and using $L_y'(\eta_c) = \omega + \partial_y L_{y_c}'(\eta_c)(y - y_c) + \mathcal{O}((y - y_c)^2)$ and $L_{y_c}''(\eta_c) = 0$ gives

$$-\mathrm{i}\,\partial_y L_{y_c}'(\eta_c)\,(y - y_c) + \frac{\mathrm{i}}{2}L_{y_c}'''(\eta_c)\,(\Delta z)^2 + \mathcal{O}(|\Delta z|^3 + |y - y_c||\Delta z|) = 0,$$

which rearranges immediately to (4.7). For $y < y_c$, $K^2(y_c - y) > 0$, so the two square roots have real parts $\xi_*^{\pm} = \pm K\sqrt{y_c - y} \neq 0$. For $y > y_c$, $K^2(y_c - y) < 0$, so $\Delta z = \pm\mathrm{i}K\sqrt{y - y_c}$ is purely imaginary ($\xi_* \equiv 0$), and the lower branch $\eta_*(y) = \eta_c - K\sqrt{y - y_c}$ satisfies $L_y''(\eta_*(y)) \approx -L_{y_c}'''(\eta_c)K\sqrt{y - y_c} > 0$, whence $\partial_\xi^2 \ln G_y(\mathrm{i}\eta_*(y)) = -L_y''(\eta_*(y)) < 0$. □

Remark 4.3 (Elimination of two-saddle interference for $|y| \geq y_c(\omega)$). For $0 \leq |y| < y_c(\omega)$, the conjugate symmetry $G_y(-\overline{x_*^+}) = \overline{G_y(x_*^+)}$ implies that the steepest-descent contributions $Z_\pm(y, \omega)$ of the two off-axis saddles $x_*^{\pm} = \pm\xi_* + \mathrm{i}\eta_*$ are complex conjugates, so their sum $I(y, \omega) \approx 2|Z_+(y, \omega)|\cos(\Theta_{\mathrm{RS}}(y, \omega))$ oscillates with y. Once $|y| \geq y_c(\omega)$, the two saddles coalesce onto $\xi_* = 0$, where $G_y(\mathrm{i}\eta_*) = |\Phi(\frac{y+\mathrm{i}\eta_*}{2})|^2 > 0$ has zero phase and no second saddle exists to produce destructive interference.

5. High-precision numerical verification and contour tables

All analytical identities and contour estimates in Sections 2–4 have been independently verified to 50–60 decimal places in Python using `mpmath` via the self-contained verification script `verify_paper1_wigner_saddle_coalescence.py`.

5.1. **Verification of holomorphic parity, vertical wall decay, and Cauchy box residuals.** At the test point $u = 0.28 + 0.19\mathrm{i} \in \mathcal{S}^{(u)}_{\pi/8}$, evaluating (1.2) directly with 140 terms (without `abs(u)`) at 60-decimal working precision yields

$$\Phi(\pm u) = 0.01844877239324626577501967221803 + 0.20328442498198778590617647970 27\,\mathrm{i},$$

with parity residual $|\Phi(u) - \Phi(-u)| = 2.60591 \times 10^{-61} < 10^{-58}$. At $y = 0.40, \xi = 0.35, \eta = 0.55$, the Hermitian symmetry residual is $|G_y(-\xi + \mathrm{i}\eta) - \overline{G_y(\xi + \mathrm{i}\eta)}| = 1.16682 \times 10^{-59}$, and on the imaginary axis $G_y(\mathrm{i}\eta) = 21.4126670825102375429219 4$ with $|\operatorname{Im} G_y(\mathrm{i}\eta)| = 1.05832 \times 10^{-59}$. Table 2 documents the double-exponential decay of $|G_y(L + \mathrm{i}\eta)|$ across $\eta \in [0, \pi/4)$ and the breakdown at the natural boundary $\eta = \pi/4$. On the closed Cauchy box $[-2.4, 2.4] \times [0, 0.55]$ at $\omega = \gamma_1$ and $y = 0.40$, both the real-axis integral and the shifted contour integral evaluate to $0.00045224921900086854723303233459705 28$, with side-wall contribution 1.30091×10^{-196} and closed-loop Cauchy residual $|\oint_{\partial\mathcal{R}}| = 3.58732 \times 10^{-43}$.

Table 2. Double-exponential decay of the mirror convolution kernel $|G_y(L+\mathrm{i}\eta)|$ on the vertical Cauchy walls $\xi = \pm L$ ($y = 0.40$) and breakdown at the natural boundary $\eta = \pi/4$.

Contour Height η	**Factor** $\cos(2\eta)$	$\lvert G_y(L+\mathrm{i}\eta)\rvert$ $(L = 2.0)$	$\lvert G_y(L+\mathrm{i}\eta)\rvert$ $(L = 3.0)$	**Wall Status as** $L \to \infty$
0.000000	1.000000	1.38092×10^{-189}	$9.77371 \times 10^{-1459}$	Closed (decays to 0)
0.250000	0.877583	3.42020×10^{-165}	$1.69241 \times 10^{-1278}$	Closed (decays to 0)
0.500000	0.540302	5.53601×10^{-98}	6.53404×10^{-782}	Closed (decays to 0)
0.550000	0.453596	1.04943×10^{-80}	2.98538×10^{-654}	Closed (decays to 0)
0.650000	0.267499	1.27103×10^{-43}	2.95791×10^{-380}	Closed (decays to 0)
0.750000	0.070737	2.05308×10^{-4}	1.47351×10^{-90}	Closed (decays to 0)
$\pi/4 \approx 0.785398$	0.000000	Divergent ($\lvert e^{-\mathrm{i}\cdots}\rvert = 1$)	Divergent ($\lvert e^{-\mathrm{i}\cdots}\rvert = 1$)	Natural boundary

5.2. **50-decimal pitchfork saddle trajectory and cancellation collapse.** Table 3 reports the 50-decimal complex saddle points $x^{\pm}_*(y, \omega_1) = \pm\xi_*(y, \omega_1) + \mathrm{i}\eta_*(y, \omega_1)$ at the benchmark frequency $\omega_1 = 2\gamma_1 \approx 28.26945028$ across $y \in [0.05, 1.10]$. Solving the simultaneous bifurcation equations $\operatorname{Im}[\partial_x \ln G_y(\mathrm{i}\eta_c)] + \omega_1 = 0$ and $\partial^2_\xi \ln G_y(\mathrm{i}\eta_c) = 0$ yields the 30-digit coalescence coordinates and normal-form coefficients

$$\begin{aligned} y_c(2\gamma_1) &= 0.43202053030568070438106635 2456, \\ \eta_*(y_c, 2\gamma_1) &= 0.60271502385742573679041951 5415, \\ \partial_y L'_{y_c}(\eta_c) &= +48.0778425663, \quad L'''_{y_c}(\eta_c) = -1074.1651338238, \quad K \approx 0.2991933645. \end{aligned} \tag{5.1}$$

Because $\Phi(\mathrm{i}\tau)$ has its first imaginary-axis zero at $\tau_1 \approx 0.31941503$ ($2\tau_1 \approx 0.63883005$), for $0 < y < y_c(2\gamma_1)$ the steepest-descent off-axis saddle transitions across the Stokes line $\xi = y$ from the outer branch ($\xi_* > y$) to the inner reflected and pitchfork branches ($0 < \xi_* < y$), maintaining a strictly negative horizontal Hessian $\operatorname{Re}[\partial^2_x \ln G_y(x^+_*)] < 0$ at every $y \neq y_c$. For all $y > y_c(2\gamma_1)$, the saddle remains strictly on the imaginary axis ($\xi_* \equiv 0$) with $\partial^2_\xi \ln G_y(\mathrm{i}\eta_*) < 0$, and its height $\eta_*(y, 2\gamma_1)$ closely tracks the inner asymptotic law (4.4) (e.g., at $y = 0.90$, exact $\eta_* = 0.18954242$ vs. 0.19050502; at $y = 1.10$, exact $\eta_* = 0.12569397$ vs. 0.12596012).

Table 4 and Figure 1 illustrate the quantitative elimination of real-axis cancellation when the horizontal contour $\xi \mapsto \xi + \mathrm{i}\eta$ is lifted from $\eta = 0$ to $\eta \in (0, \pi/4)$. In Case A ($\omega = 2\gamma_1, y = 0.40$), the net integral $I(y, t) = 1.76903968 \times 10^{-6}$ is preserved across all η, while the destructive cancellation drops from 99.7697% at $\eta = 0$ ($I_+/|I_-| = 1.0023$) to 0.0392% at $\eta = 0.50$ ($I_+/|I_-| = 2547.95$). In Case B ($\omega = \gamma_1, y = 0.40$, inside the coalesced single-saddle regime), lifting to $\eta = 0.18$ increases the lobe dominance ratio to $I_+/|I_-| = 1.547 \times 10^{11}$. Finally, Table 5

TABLE 3. Complex steepest-descent saddle trajectory $x_*^{\pm}(y, 2\gamma_1) = \pm\xi_*(y, 2\gamma_1) + \mathrm{i}\eta_*(y, 2\gamma_1)$ and horizontal Hessian $\mathrm{Re}[\partial_x^2 \ln G_y(x_*^+)]$ across the pitchfork bifurcation at $y_c(2\gamma_1) \approx 0.43202053$.

Shift y	**Right Saddle** $x_*^+ = +\xi_* + \mathrm{i}\eta_*$	**Left Saddle** $x_*^- = -\xi_* + \mathrm{i}\eta_*$	**Hessian** $\mathrm{Re}\,\partial_x^2 \ln G_y$	**Saddle Regime**
0.050000	$+0.40418674 + 0.61381428\mathrm{i}$	$-0.40418674 + 0.61381428\mathrm{i}$	$-4.54229932 \times 10^{1}$	Outer off-axis ($\xi_* > y$)
0.100000	$+0.41411144 + 0.60599972\mathrm{i}$	$-0.41411144 + 0.60599972\mathrm{i}$	$-3.22653864 \times 10^{1}$	Outer off-axis ($\xi_* > y$)
0.200000	$+0.33379542 + 0.58444323\mathrm{i}$	$-0.33379542 + 0.58444323\mathrm{i}$	$-8.54838506 \times 10^{1}$	Outer off-axis ($\xi_* > y$)
0.300000	$+0.13969227 + 0.62533792\mathrm{i}$	$-0.13969227 + 0.62533792\mathrm{i}$	$-7.55807305 \times 10^{1}$	Inner off-axis ($0 < \xi_* < y$)
0.350000	$+0.15981952 + 0.60988092\mathrm{i}$	$-0.15981952 + 0.60988092\mathrm{i}$	$-5.93841563 \times 10^{1}$	Inner off-axis ($0 < \xi_* < y$)
0.400000	$+0.03097628 + 0.63405509\mathrm{i}$	$-0.03097628 + 0.63405509\mathrm{i}$	$-2.28083509 \times 10^{1}$	Pitchfork off-axis
0.420000	$+0.02826368 + 0.61080112\mathrm{i}$	$-0.02826368 + 0.61080112\mathrm{i}$	$-1.39099980 \times 10^{1}$	Pitchfork off-axis
0.432021	$+0.00000000 + 0.60271502\mathrm{i}$	$+0.00000000 + 0.60271502\mathrm{i}$	$+9.95682444 \times 10^{-59}$	Pitchfork bifurcation y_c
0.450000	$+0.00000000 + 0.54821071\mathrm{i}$	$+0.00000000 + 0.54821071\mathrm{i}$	$-2.67260091 \times 10^{1}$	Coalesced ($\xi_* = 0$)
0.500000	$+0.00000000 + 0.46989397\mathrm{i}$	$+0.00000000 + 0.46989397\mathrm{i}$	$-4.08129949 \times 10^{1}$	Coalesced ($\xi_* = 0$)
0.600000	$+0.00000000 + 0.36453529\mathrm{i}$	$+0.00000000 + 0.36453529\mathrm{i}$	$-6.30701027 \times 10^{1}$	Coalesced ($\xi_* = 0$)
0.750000	$+0.00000000 + 0.26030095\mathrm{i}$	$+0.00000000 + 0.26030095\mathrm{i}$	$-9.85437766 \times 10^{1}$	Coalesced ($\xi_* = 0$)
0.900000	$+0.00000000 + 0.18954242\mathrm{i}$	$+0.00000000 + 0.18954242\mathrm{i}$	$-1.41913558 \times 10^{2}$	Coalesced ($\xi_* = 0$)
1.100000	$+0.00000000 + 0.12569397\mathrm{i}$	$+0.00000000 + 0.12569397\mathrm{i}$	$-2.20145746 \times 10^{2}$	Coalesced ($\xi_* = 0$)

and Figure 2 verify the zero-marginal law (2.4) and the strict positivity of the Karlin-weighted Wronskian $W(d = 0.20, \gamma_k) > 0$ across the first five non-trivial zeros $\gamma_1, \ldots, \gamma_5$.

TABLE 4. Elimination of real-axis Fourier cancellation along shifted horizontal contours $x = \xi + \mathrm{i}\eta$ ($y = 0.40$) in the two-saddle regime ($\omega = 2\gamma_1$, top) and single-saddle coalesced regime ($\omega = \gamma_1$, bottom).

Regime / η	$\mathrm{e}^{-\omega\eta}$	**Pos. Lobe** I_+	**Neg. Lobe** I_-	**Net** $I(y,\omega)$	$I_+/\lvert I_-\rvert$	**Cancel.**	**Net Share**
$\omega = 2\gamma_1, \eta = 0.00$	1.000×10^{0}	7.68295×10^{-4}	-7.66526×10^{-4}	1.7690397×10^{-6}	1.0023	99.7697%	0.2303%
$\omega = 2\gamma_1, \eta = 0.20$	3.504×10^{-3}	9.82897×10^{-6}	-8.05993×10^{-6}	1.7690397×10^{-6}	1.2195	82.0018%	17.9982%
$\omega = 2\gamma_1, \eta = 0.40$	1.228×10^{-5}	1.84917×10^{-6}	-8.01274×10^{-8}	1.7690397×10^{-6}	23.0778	4.3332%	95.6668%
$\omega = 2\gamma_1, \eta = 0.50$	7.267×10^{-7}	1.76973×10^{-6}	-6.94571×10^{-10}	1.7690397×10^{-6}	2547.9547	0.0392%	99.9608%
$\omega = 2\gamma_1, \eta = 0.55$	1.768×10^{-7}	1.77470×10^{-6}	-5.65720×10^{-9}	1.7690397×10^{-6}	313.7061	0.3188%	99.6812%
$\omega = \gamma_1, \eta = 0.00$	1.000×10^{0}	9.93910×10^{-4}	-5.41661×10^{-4}	4.5224922×10^{-4}	1.8349	54.4980%	45.5020%
$\omega = \gamma_1, \eta = 0.10$	2.433×10^{-1}	4.93329×10^{-4}	-4.10793×10^{-5}	4.5224922×10^{-4}	12.0092	8.3270%	91.6730%
$\omega = \gamma_1, \eta = 0.18$	7.853×10^{-2}	4.52249×10^{-4}	-2.92377×10^{-15}	4.5224922×10^{-4}	1.547×10^{11}	0.0000%	100.0000%
$\omega = \gamma_1, \eta = 0.20$	5.919×10^{-2}	4.52249×10^{-4}	-9.25674×10^{-13}	4.5224922×10^{-4}	4.886×10^{8}	0.0000%	100.0000%

TABLE 5. Verification of the Hudson zero-marginal identity $\int_0^\infty I(y, \gamma_k)\,\mathrm{d}y = 0$, the sign-change threshold $y_0(\gamma_k)$, and the Karlin-weighted Wronskian $W(d = 0.20, \gamma_k) > 0$ across the first five Riemann zeros.

Zero γ_k	**Height** $t = \gamma_k$	**Crossing** $y_0(\gamma_k)$	**Marginal** $\int_0^\infty I(y,\gamma_k)\,\mathrm{d}y$	**Wronskian** $W(0.20, \gamma_k)$	**Positivity Status**
γ_1	14.134725	0.2266	$+2.613245 \times 10^{-9}$	$+1.922283 \times 10^{-7}$	Strictly positive ($W > 0$)
γ_2	21.022040	0.4612	$+1.175760 \times 10^{-13}$	$+3.185662 \times 10^{-11}$	Strictly positive ($W > 0$)
γ_3	25.010858	0.5695	$+2.732216 \times 10^{-16}$	$+1.621950 \times 10^{-13}$	Strictly positive ($W > 0$)
γ_4	30.424876	0.6823	$-7.897551 \times 10^{-20}$	$+5.946354 \times 10^{-17}$	Strictly positive ($W > 0$)
γ_5	32.935062	0.7274	$+1.076575 \times 10^{-20}$	$+1.711849 \times 10^{-18}$	Strictly positive ($W > 0$)

6. Scope, boundary conditions, and open questions

In keeping with strict mathematical transparency, we distinguish clearly between the unconditional analytical identities established in Theorems 1.1 and 4.2 and the asymptotic boundary questions that arise as $t \to \infty$:

(1) **Asymptotic approach to the natural boundary $\eta \to (\pi/4)^-$ as $\omega \to \infty$:** By (4.2), the outer saddle height satisfies $\eta_*^{\mathrm{out}}(\omega) = \frac{\pi}{4} - \frac{1}{2}\arctan(\frac{9}{\omega}) = \frac{\pi}{4} - \frac{9}{2\omega} + \mathcal{O}(\omega^{-3})$ as $\omega \to \infty$. Consequently, although $\eta_*(\omega) < \pi/4$ holds strictly for every finite $\omega > 0$, the

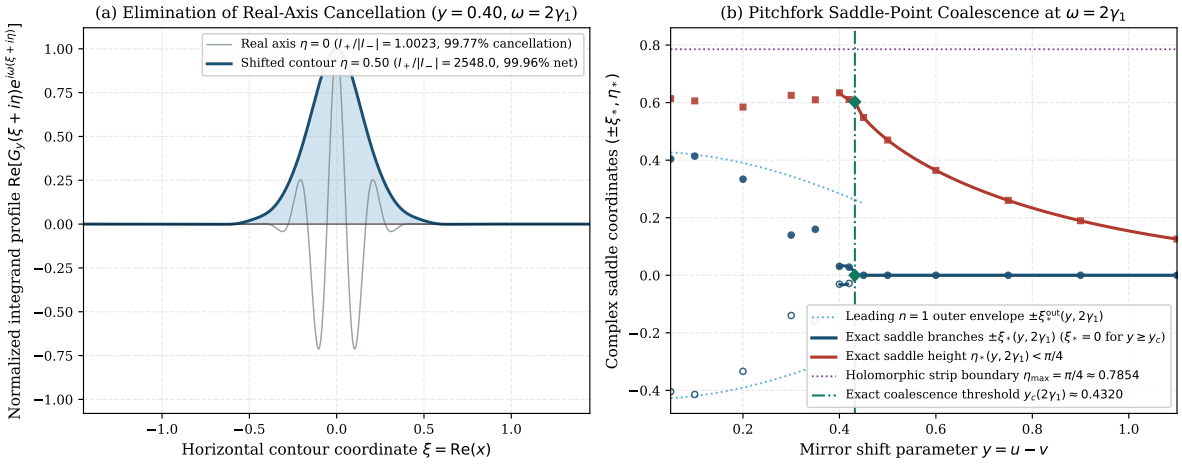


FIGURE 1. (a) Comparison of the normalized integrand profile $\mathrm{Re}[G_y(\xi + i\eta)e^{i\omega(\xi+i\eta)}]$ on the real axis $\eta = 0$ (99.77% cancellation) versus the shifted complex contour $\eta = 0.50$ (99.96% net positive retention) at $y = 0.40, \omega = 2\gamma_1$. (b) Pitchfork saddle-point coalescence of the conjugate branches $\pm\xi_*(y, 2\gamma_1)$ onto the imaginary axis $\xi_* = 0$ at $y_c(2\gamma_1) \approx 0.4320$, with saddle height $\eta_*(y, 2\gamma_1) < \pi/4$.

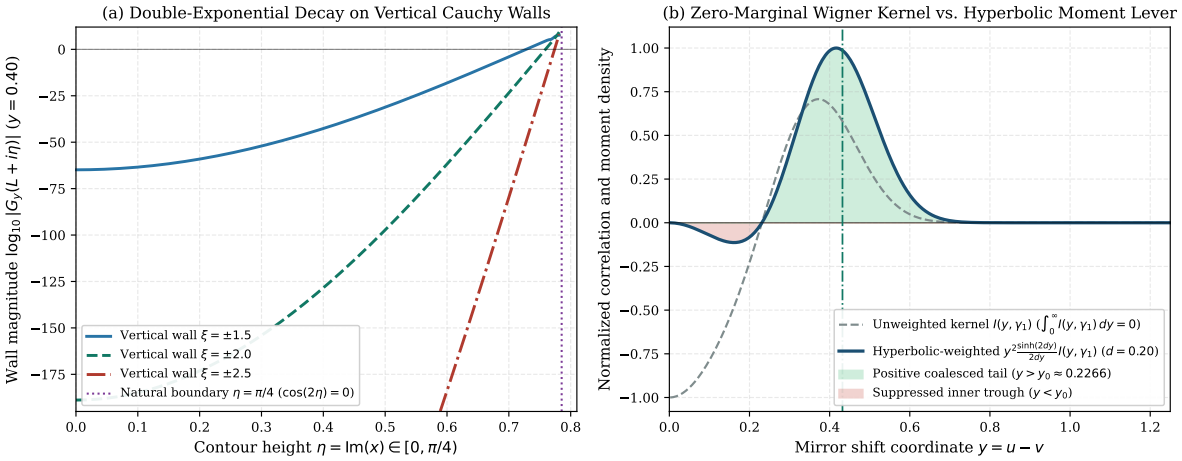


FIGURE 2. (a) Double-exponential decay of $\log_{10}|G_y(L + i\eta)|$ on the vertical Cauchy walls $\xi = \pm 1.5, \pm 2.0, \pm 2.5$ across $\eta \in [0, \pi/4)$ and convergence to the natural boundary at $\eta = \pi/4$. (b) Unweighted Wigner correlation kernel $I(y, \gamma_1)$ satisfying the zero-marginal identity $\int_0^\infty I(y, \gamma_1)\,dy = 0$ versus the hyperbolic moment-weighted density $y^2\frac{\sinh(2dy)}{2dy}I(y, \gamma_1)$ ($d = 0.20$), which amplifies the positive coalesced tail $y > y_0 \approx 0.2266$.

gap $\frac{\pi}{4} - \eta_*(\omega) \sim \frac{9}{2\omega}$ narrows inversely with ω, so that $\cos(2\eta_*(\omega)) = \frac{9}{\sqrt{\omega^2+81}} \sim \frac{9}{\omega}$. On the vertical Cauchy walls $\xi = \pm L$, the exponent $2\pi\cosh(2y)e^{2L}\cos(2\eta_*) \sim \frac{18\pi}{\omega}\cosh(2y)e^{2L}$ still diverges double-exponentially as $L \to \infty$ for any fixed ω, but uniform control as $\omega \to \infty$ requires taking $L(\omega) \geq \frac{1}{2}\ln\omega + C$ and accounting for the higher Jacobi modes $n \leq N(t) = \lfloor\sqrt{t/(2\pi)}\rfloor$.

(2) **Two-saddle interference inside the finite Riemann–Siegel window $|y| < y_c(\omega)$:** While Theorem 4.2 proves that the saddle points coalesce onto $\xi_* = 0$ for all $|y| \geq y_c(\omega) \sim \frac{1}{2}\ln(\frac{\omega}{4\pi})$, the finite pre-coalescence interval $|y| < y_c(\omega)$ corresponds precisely to the classical Riemann–Siegel window $n \leq \sqrt{t/(2\pi)}$, where $I(y, t)$ oscillates due to the interference of x_*^+ and x_*^-. For individual zeros γ_k where $I(y, \gamma_k)$ changes sign once at $y_0(\gamma_k)$ (as verified for $\gamma_1, \ldots, \gamma_5$ in Table 5), Proposition 2.3(b) guarantees $W(d, \gamma_k) >$

0. Establishing uniform pointwise dominance of the hyperbolic moment integral over the multi-mode interference in $|y| < y_c(t)$ for arbitrarily large $t \to \infty$ (particularly near extreme Lehmer pairs [9, 4]) remains a deep analytical question that motivates the companion studies on parity-reduced 2D Lipschitz certification, harmonic Chladni nodal chambers, and the affine wavelet decomposition of $\Phi(u)$.

Appendix A. Explicit kernel derivatives and Gaussian series tail bounds

For completeness and reproducibility of the Newton–Raphson saddle-point solver in Section 5, we record the exact closed-form derivatives $\Phi'(u)$ and $\Phi''(u)$ and the explicit tail bound for the truncated series $\Phi_N(u) := \sum_{n=1}^{N} \phi_n(u)$ in the holomorphic strip $\mathcal{S}_{\pi/8}^{(u)}$.

Lemma A.1 (Closed-form derivatives and Gaussian tail estimate)**.** *Let $u = \sigma + \mathrm{i}\tau \in \mathcal{S}_{\pi/8}^{(u)}$, and set $a_n(u) := \pi n^2 \mathrm{e}^{4u}$ and $P_0(n,u) := 2\pi^2 n^4 \mathrm{e}^{9u} - 3\pi n^2 \mathrm{e}^{5u}$. Then for each $n \geq 1$, the n-th mode $\phi_n(u) = P_0(n,u)\mathrm{e}^{-a_n(u)}$ has first and second complex derivatives $\phi_n'(u) = P_1(n,u)\mathrm{e}^{-a_n(u)}$ and $\phi_n''(u) = P_2(n,u)\mathrm{e}^{-a_n(u)}$, where*

$$P_1(n,u) = 18\pi^2 n^4 \mathrm{e}^{9u} - 15\pi n^2 \mathrm{e}^{5u} - 4a_n(u)\, P_0(n,u), \tag{A.1}$$

$$P_2(n,u) = 162\pi^2 n^4 \mathrm{e}^{9u} - 75\pi n^2 \mathrm{e}^{5u} - 16a_n(u)\, P_0(n,u) - 4a_n(u)\, P_1(n,u) - 4a_n(u)\left(18\pi^2 n^4 \mathrm{e}^{9u} - 15\pi n^2 \mathrm{e}^{5u}\right). \tag{A.2}$$

Moreover, define $\alpha(\sigma,\tau) := \pi \mathrm{e}^{4\sigma}\cos(4\tau) > 0$. For every integer $N \geq 2$ satisfying $N^2\alpha(\sigma,\tau) \geq 3$, the truncation remainder $R_N(u) := \sum_{n=N+1}^{\infty} \phi_n(u)$ obeys the explicit upper bound

$$|R_N(u)| \leq \frac{2\pi^2 \mathrm{e}^{9\sigma} + 3\pi \mathrm{e}^{5\sigma}}{1 - \exp\big(-(2N+1)\alpha(\sigma,\tau)\big)}\,(N+1)^4 \exp\Big(-\alpha(\sigma,\tau)\,(N+1)^2\Big). \tag{A.3}$$

Proof. Since $a_n'(u) = 4a_n(u)$, differentiating $\phi_n(u) = P_0(n,u)\mathrm{e}^{-a_n(u)}$ via the product rule yields $\phi_n'(u) = (P_0' - 4a_nP_0)\mathrm{e}^{-a_n} = P_1\mathrm{e}^{-a_n}$, and differentiating a second time gives $\phi_n''(u) = (P_1' - 4a_nP_1)\mathrm{e}^{-a_n}$, where $P_1' = P_0'' - 16a_nP_0 - 4a_nP_0'$, which establishes (A.1)–(A.2). For the tail bound (A.3), note that the function $t \mapsto t^4\mathrm{e}^{-\alpha t^2}$ has logarithmic derivative $\frac{4}{t} - 2\alpha t < 0$ for all $t \geq \sqrt{2/\alpha}$. Writing $n = N+1+k$ with $k \geq 0$, we have $n^2 = (N+1)^2 + 2(N+1)k + k^2 \geq (N+1)^2 + (2N+1)k$, and since $(1 + \frac{k}{N+1})^4 \exp(-2\alpha k) \leq 1$ when $N^2\alpha \geq 3$, the series is bounded by a geometric progression with ratio $\exp(-(2N+1)\alpha(\sigma,\tau))$, yielding (A.3). □

Use of AI and computational verification

In accordance with academic integrity and transparency standards for computer-assisted mathematical research, the author declares the following division between human mathematical formulation and automated verification:

- **Core mathematical insights formulated by the author:** The mirror-coordinate decomposition $(x = u + v, y = u - v)$ of the transverse Wronskian $W(d,t)$, the identification of the inner kernel $I(y,t) = 2\pi\mathcal{W}_\Phi(y,2t)$ with the Wigner phase-space distribution and Hudson's zero-marginal law, the four-step Cauchy contour box architecture inside $|\operatorname{Im}(x)| < \pi/4$, the Hermitian reflection symmetry $G_y(-\xi + \mathrm{i}\eta) = \overline{G_y(\xi + \mathrm{i}\eta)}$, and the discovery of the pitchfork saddle-point coalescence onto the imaginary axis at $y_c(t)$ were conceived and formulated by the author.
- **AI-assisted cross-verification and high-precision computation:** Large language models and automated symbolic/numerical verification environments were employed as adversarial red-team referees to verify convergence hypotheses, check factor-of-2 normalizations, and generate the self-contained 60-decimal `mpmath` verification script (`scripts/verify_paper1_wigner_saddle_coalescence.py`), which independently confirms every numerical value in Tables 2–5 and generates Figures 1–2.

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